ION-Sim: A Novel Open-Source Simulation Framework for Intraoperative Neurophysiological Monitoring"

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Database Organization (MySql)

This section describes the organization and use of the database that allows transparent management of simulation activities and learning processes.

The database structure contains different modules (relational tables) that contain data structures that allow storing information to identify the user ("sorcerer's apprentice"), information describing the simulation or learning work sessions performed, and information on the results obtained in the learning sessions to build a report on the activities performed.

The database contains a further series of tables that describe the different simulation and learning scenarios. The structure of the scenarios is unimodal and can thus be combined into more complex multimodal scenarios that realistically describe intraoperative conditions.

Teaching activities are carried out at different levels of complexity following a gradual process, in relation to the accumulated results. In general, learning of individual neurophysiological test modalities is proposed, anomalies that must be captured and interpreted by the learner are proposed, and realistic surgical scenarios are proposed in which the best approach to neurophysiological monitoring is required.

Tabella "subjects"

Description of the user information table **"subjects"** (fields present):

1. "*firstname*" : surname
2. "*lastname*" : name
3. "*age*" : age
4. "*gender*" : gender
5. "*datetime*" : a field that automatically saves the timestamp of the new user's insertion
6. "*dbaseclass*" : describes the type of learning approach
    - Simulation = ION-Sim
    - Learning of single modalities = UNI
    - Multimodal learning = MULTI
    - Generation of advanced scenarios = ADVSCENE

- “Level” : contains the learning modality that the user chooses at the beginning of the training and is automatically updated based on the results obtained from the learning sessions.

Name: subjects

Comment: list of subjects recorded (uniques): le informazioni identificative utilizzate allo scopo di identificare lo stesso operatore in sessioni differenti e successive. Il campo di baseclass può essere utilizzato in seguito per raggruppare diverse tipologie di utenti: discenti, insegnanti, operatore autorizzato a inserire nuovi scenari o anomalie ecc...

Columns: Add Remove Up Down

#	Name	Datatype	Length/Set	Unsigned	Allow NULL	Zero...	Default	Cor
1	idcode	INT	11	☐	☐	☐	AUTO_INCREMENT	
2	firstname	VARCHAR	50	☐	☑	☐	NULL	
3	lastname	VARCHAR	50	☐	☑	☐	NULL	
4	height	INT	11	☐	☑	☐	NULL	
5	weight	INT	11	☐	☑	☐	NULL	
6	age	INT	11	☐	☑	☐	NULL	
7	gender	CHAR	1	☐	☑	☐	NULL	
8	fname	VARCHAR	50	☐	☑	☐	NULL	
9	dbaseclass	VARCHAR	50	☐	☑	☐	NULL	
10	level	VARCHAR	20	☐	☐	☐	No default	
11	datetime	TIMESTAMP		☐	☑	☐	NULL	

"sessions" Table

Description of the table of sessions held **"sessions"** (fields present) with a one-to-many relationship with the "subjects" table:

1. *“session”* : identifies the type of protocol to simulate or interact with (the list of protocols is stored in a specific "scenari_setup" table)
2. *“taskname”*: identification code of the protocol chosen in "session" (always from the "scenari_setup" table)
3. *“sesnote”* : free annotations that can be inserted in the initial function of choosing the protocol to use but can be updated at any time and are strictly correlated to the selected session..
4. *“datetime”* : a field that automatically saves the timestamp of the new session's insertion.

Name: sessions

Comment: sessions (multiples but unique indexed) [one to many] recorded for each operator: multipe sessions for each operator.Il campo dbasename viene aggiornato dalla scelta del campo dbaseclass della tabella soggetti.

Columns: Add Remove Up Down

#	Name	Datatype	Length/Set	Unsigned	Allow NULL	Zero...	Default	Con
1	idcode	INT	11	☐	☑	☐	NULL	
2	idsession	INT	11	☐	☐	☐	AUTO_INCREMENT	
3	session	VARCHAR	50	☐	☑	☐	NULL	
4	taskname	VARCHAR	50	☐	☑	☐	NULL	
5	simulacode	CHAR	50	☐	☑	☐	NULL	
6	sesnote	TEXT		☐	☑	☐	NULL	
7	fname	VARCHAR	50	☐	☑	☐	NULL	
8	dbasename	VARCHAR	50	☐	☑	☐	NULL	
9	datetime	TIMESTAMP		☐	☑	☐	NULL	

Tabella "scenari_setup"

Description of the table listing the set protocols **"scenari_setup"** (fields present):

1. *"nome_scenario"* : protocol description
2. *"label_id"* : label of maximum 5 characters that identifies the unique protocol
3. *"descrizione"* : description of the characterizing elements of the protocol
4. *"trigger_tipo"* : encodes the type of stimulation present in the selected protocol

Below is the list of scenarios inserted and for which the simulation module is available. As can be seen, each scenario corresponds to a single simulation modality, and different simulation modalities (multimodal scenario) can be conveyed into the IOM scenario to represent a true realistic intraoperative scenario.

Name: scenari_setup

Comment: descrive tutti gli scenari che sono presenti nei modelli di simulazione e che possono essere selezionati dal discente per apprendimento o approfondimento. Attualmente 17 scenari unimodali e uno (IOM) multimodale. Si disporrano diversi scenari multimodali a seconda della richista del fruitore.

Columns: Add Remove Up Down

#	Name	Datatype	Length/Set	Unsigned	Allow NULL	Zero...	Default	Comment
1	scenario_id	INT	11	☐	☐	☐	AUTO_INCREME..	
2	label_id	CHAR	20	☐	☐	☐	No default	
3	nome_scenario	VARCHAR	100	☐	☐	☐	No default	
4	descrizione	TEXT		☐	☑	☐	NULL	
5	trigger_tipo	VARCHAR	50	☐	☐	☐	No default	
6	trigger_delay	INT	11	☐	☑	☐	NULL	
7	data_creazione	TIMESTAMP		☐	☐	☐	current_timesta...	

IOM_Simula.scenari_setup: 18 rows total (exact) Next Show all Sorting (1) Columns (7/7) Filter

#	scenario_id	label_id	nome_scenario	descrizione	trigger_tipo	trigger_delay
1	9	BAEP	baep	baep brain stem acoustic evoked potentials	stim_acoustic	(NULL)
2	15	CCEP	cortico-cortical-evoked_potentials	ccep cortico cortical evoked potentials	stim	(NULL)
3	10	DCScn	dcs_cranial_nerves	direct stimulation of cranial nerves	stim	(NULL)
4	11	DCSsp	dcs_spinal_cord	direct stimulation of spinal cord: posterior cords	stim	(NULL)
5	13	EEG	eeg_spontaneous	eeg signal with artifacts	none	(NULL)
6	14	EcoG	electrocort_spont	ecog from cortical or sub cortical electrodes	none	(NULL)
7	17	EMG	emg_arm_limb	emg free running from limb muscles	none	(NULL)
8	16	EMGcn	emg_cranial_nerves	emg free running from muscles innervated by crania...	none	(NULL)
9	18	EMGtr	emg_triggered	emg triggered by stimulus, alias single mep response	stim	(NULL)
10	7	IOM	iom_multimodal	multimodal monitoring	stim_multi	(NULL)
11	6	MEPai	mep_arti_inferiori	lower limbs motor responses by transcranial stimulation	stim	(NULL)
12	5	MEPas	mep_arti_superiori	upper limbs motor responses by transcranial stimulati...	stim	(NULL)
13	4	MEPcb	mep_cranial_nerves	mep recorded from [target muscles] innervated by cr...	stim	10
14	12	MEPdw	mep_dwave	D-Wave obtained from descending volley of pyramida...	stim	(NULL)
15	1	CAP	nervo_periferico	compound action potential - afferent	stim	10
16	3	SEPAI	sep_arti_inferiori	sep lower limbs due stimulation of tibial nerve at ankle	stim	10
17	2	SEPAS	sep_arti_sup	sep upperlimb due to stimulation of median nerve at ...	stim	10
18	8	VEP	vep-flash	visul evoked potential due to flash retinal stimulation	stim_flash	(NULL)

"scenari_anomaly" Table

Description of the table listing the type of anomalies **"scenari_anomaly"**, activatable for each protocol identified in the "scenari_setup" table with a one-to-many relationship:

1. *"label_id"* : mnemonic identifier of the type of anomaly associable to the specific protocol (all anomaly details are managed within the associated json file)
2. *"descrizione"* : synthetic description of the type of anomaly that can be inserted into the specific protocol chosen in the "scenari_setup" table
3. *"json_anomaly_id"* : the filename (managed transparently by the system) to store all the parameters necessary to inject the anomaly defined in the chosen protocol.
4. *"data_creazione"* : creation date of the anomaly type. An advanced user can be authorized to create and insert new anomalies for each specific protocol.

Note that the .json file contains the indications for executing or injecting a specific anomaly in a given protocol.

Name: scenari_anomaly

Comment: descrive le tipologie di anomalie che si possono verificare per ciascuno scenario descritto nella tabella scenari_setup. Tutti i dati relativi alle impostazioni di ciascuna anomalia e le modalità operative sono descritti in un file .json a cui la tabella fa riferimento e salvati nella directory "anomaly"

Columns: Add Remove Up Down

#	Name	Datatype	Length/Set	Unsigned	Allow NULL	Zero...	Default	Comme
1	anomaly_id	INT	11	☐	☐	☐	AUTO_INCREME..	
2	scenario_id	INT	11	☐	☐	☐	No default	
3	label_id	CHAR	20	☐	☐	☐	No default	
4	descrizione	TEXT		☐	☑	☐	NULL	
5	json_anomaly_id	VARCHAR	20	☐	☐	☐	No default	
6	data_creazione	TIMESTAMP		☐	☐	☐	current_timesta...	

The list of anomaly conditions that can be tested is certainly more extensive than what is exemplified below but provides an idea of how this functionality has been implemented.

IOM_Simula.scenari_anomaly: 28 rows total (exact) Next Show all Sorting (1) Colum

#	anomaly_id	scenario_id	label_id	descrizione	json_anomaly_id
1	1	1	CAP_1	CAP reduced amplitude: loss of fibers = block	anomaly/cap_1.json
2	8	1	CAP_2	CAP increased latency: partial demyelination = slowi...	anomaly/cap_2.json
3	6	2	SEPAS_1	sepas decrease N20 amplitude: central cause	anomaly/sepas_1.json
4	7	2	SEPAS_2	sepas decrease N20 amplitude: reduced stimuls inte...	anomaly/sepas_2.json
5	9	2	SEPAS_3	sepas decrease N20 amplitude bilaterally: systemic c...	anomaly/sepas_3.json
6	10	3	SEPAI_1	sepai decrease P37 amplitude: central cause	anomaly/sepai_1.json
7	14	4	MEPCB_1	mepcb with absent muscles responses	anomaly/mepcb_1.json
8	22	4	MEPCB_1	mep cranial muscles transcranial pulse: no response	anomaly/mepcb_1.json
9	21	5	MEPAS_1	mep upper limb transcranial train pulse: no response	anomaly/mepas_1.json
10	20	6	MEPAI_1	mep lower limb transcranial train pulse: no response	anomaly/mepai_1.json
11	2	7	IOM_1	iom sovratentoriale	anomaly/iom_1.json
12	3	7	IOM_2	iom sottotentoriale	anomaly/iom_2.json
13	4	7	IOM_3	iom spinale	anomaly/iom_3.json
14	15	7	IOM_4	iom cranial nerves monitor	anomaly/iom_4.json
15	25	8	VEP_1	vep da flash: dx response absent, sn regular	anomaly/vep_1.json
16	26	8	VEP_2	vep flash: dx response regular, sn absent	anomaly/vep_2.json
17	5	9	BAEP_1	baep latency increase V wave	anomaly/bape_1.json
18	27	10	DCSCN_1	dcs cranial nerves single stimulus: no response	anomaly/dcscn_1.json
19	28	11	DCSSP_1	dcs peripheral nerve: conduction block	anomaly/dcssp_1.json
20	23	12	MEPDW_1	D-wave transcranialpulse: no response from spinal co...	anomaly/mepdw_1.json
21	24	12	MEPDW_2	D-wave transcranial pulse: no response from spinal c...	anomaly/mepdw_2.json
22	11	13	EEG_1	eeg diffuse slowing (theta- delta) in al channels	anomaly/eeg_1.json
23	12	13	EEG_2	eeg diffuse attenuation of amplitude in all channels	anomaly/eeg_2.json
24	17	14	ECOG_1	ecog spike with focal pattern	anomaly/ecog_1.json
25	16	15	CCEP_1	ccep cortico -cortical evoked potentials loss	anomaly/ccep_1.json
26	18	16	EMGCN_1	emg cranial nerves spontaneosu discharge C-type	anomaly/emgcn_1.json
27	13	17	EMG_1	emg with intermittent bursts of spontaneous activity	anomaly/emg_1.json
28	19	18	EMGTR_1	emg triggered by stimulus : block 50%	anomaly/emgtr_1.json

Note that in the json file describing the characteristics of the anomaly to be tested, typical descriptive keys of the specific session identified by the taskname are inserted.

"sessions_decisions" Table

Description of the "**sessions_decisions**" table contains the results of the simulations and learning performed in each session, with a one-to-one relationship with the "sessions" table. It contains the following fields:

1. "*timestamp_decision*" : contains the date and time of the response execution
2. "*azione_presa*" : contains the string value of the response provided
3. "*valore_parametro*" : additional parameter correlated to the response
4. "*esito-corretto*" : contains one or zero depending on whether the response is correct or incorrect
5. "*punti_ottenuti*" : score accumulated for the test performed in the session
6. "*report_file*" : the filename of the report associated with the execution of the specific session

Name: session_decisions

Comment: sessions_decisions recorded and operated by every subject (one to one) during a specific session. Contains the answers (azione_presa), a parameter, a value for correct or incorrect resppnse and points obtained. Also is present the filename of attached report on session executed.

Columns: Add Remove Up Down

#	Name	Datatype	Length/Set	Unsigned	Allow NULL	Zero...	Default	Comme
1	decision_id	INT	11	☐	☐	☐	AUTO_INCREME..	
2	idsession	INT	11	☐	☐	☐	No default	
3	timestamp_decision	TIMESTAMP		☐	☐	☐	current_timesta...	
4	azione_presa	VARCHAR	100	☐	☐	☐	No default	
5	valore_parametro	DECIMAL	8,2	☐	☑	☐	NULL	
6	esito_corretto	TINYINT	1	☐	☐	☐	No default	
7	punti_ottenuti	INT	11	☐	☑	☐	'0'	
8	report_file	VARCHAR	50	☐	☑	☐	NULL	

The development of the application involves the production of a final report, summarizing the learning capabilities and the results obtained in the different sessions held, producing an index of knowledge advancement and acquired skills.